

Question	Answer	Marks	Guidance
6(a)	Most of the α -particles pass <u>straight</u> through (the atom) OR Most of the α -particles have (very) small deviations	B1	Allow 'majority' / 'nearly all' / 'large proportion' for 'most' Allow 'particles' / 'them' to mean α-particles (BOD) Allow 'no deviation' / 'undeviated' / 'deviate by small / insignificant / minimal angles' instead of small deviations Allow 'deflection' instead of deviation Allow 'gold / leaf' for atom Ignore 'some' instead of 'most' Ignore 'a large <u>number</u>' as this is not comparative Ignore just 'pass through' for 'pass straight through' Ignore 'only a (very) small proportion of alpha particles had (very) large deviations' as this does not necessarily mean the rest had little or no deviation
6(b)(i)	$(p =) mv$	C1	Allow any mass $\times$ any velocity e.g. $4(u) \times 5.60 \times 10^6$ or $1.66 \times 10^{-27} \times 5.60 \times 10^6$ or $1.67 \times 10^{-27} \times 5.60 \times 10^6$
	$= 4 \times 1.66 \times 10^{-27} \times 5.60 \times 10^6 = 3.72 \times 10^{-20} \text{ (kg m s}^{-1}\text{)}$	A1	As this is a 'show that' question the full working and answer must be seen Not 1.67×10^{-27} or 1.6×10^{-27} for A1 mark

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6(b)(ii)	$4(u) \times 5.60 \times 10^6 = 197(u)v - 4(u) \times 5.38 \times 10^6$ $3.72 \times 10^{-20} = (197 \times 1.66 \times 10^{-27}v) - (4 \times 1.66 \times 10^{-27} \times 5.38 \times 10^6)$	C1	If u is present in equation it must be present in all terms Allow 1.67 instead of 1.66 as gives correct answer to 2SF
	speed = $2.2 \times 10^5 \text{ m s}^{-1}$	A1	<u>Special case:</u> $4(u) \times 5.60 \times 10^6 = 197(u)v + (4(u) \times 5.38 \times 10^6)$ leading to 4500 m s^{-1} credit 1/2 working must be seen <u>Alternative method</u> if candidates (correctly) assume elastic collision and use speed of approach = speed of separation $\text{speed} = 5.60 \times 10^6 - 5.38 \times 10^6 \text{ (C1)}$ $= 2.2 \times 10^5 \text{ (A1)}$
6(b)(iii)	(absolute uncertainty =) $5 \times 10^4 + 5 \times 10^4$ $= 1 \times 10^5$	C1	Allow $2 \times 5 \times 10^4 = 1 \times 10^5$ Allow Δv to mean absolute uncertainty in v
	percentage uncertainty = $(1 \times 10^5 / (5.60 \times 10^6 + 5.38 \times 10^6)) \times 100$ $= 0.9\%$	A1	Allow answers to 2 or more SF that round to 0.9
6(c)	3	A1	Allow '3 up' or '3u'
6(d)	One quark and one antiquark	B1	Allow 'quark and antiquark' (assume 1 of each) Allow q $\bar{q}$ BOD Ignore named quarks whether correct or incorrect e.g. 'up quark and antidown quark' as general composition required Ignore '2 quarks'